BRADLEY
University

# CS 330: Net-Centric Computing – Spring 2024

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## Lab 2: TCP, UDP, IP

**Instructions:**

- Wireshark traces for this lab can be found in the `"Lab2-traces.zip"` file on Canvas.
- Required screenshots, indicated by [screenshot], must be added as evidence to receive full credit. Adding screenshots elsewhere is optional but encouraged.

| Question: | 1 | 2 | 3 | Total |
|---|---|---|---|---|
| Points: | 10 | 10 | 10 | 30 |

## Part I: TCP

**Q.1** Load the file `"tcp-1.pcapng"` in Wireshark and answer the following questions. The trace file was captured while uploading '1600.txt' file from a computer (`10.63.7.192`) to `gaia.cs.umass.edu` web server (`128.119.245.12`) using the HTTP POST method.

(a) What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and `gaia.cs.umass.edu` web server? [2]

(b) What are the sequence number and acknowledgement number of the first SYNACK packet sent from the server to the client? How were the values determined by the server? [2]

*[Note: relative* `seq` *and* `ack` *values displayed by Wireshark are sufficient, actual values are not required.]*

(c) What are the minimum and maximum amount of available buffer spaces advertised at the receiver for the entire trace? [screenshot] [2]

(d) Are there any re-transmitted segments in the trace file? Clearly explain why you said yes or no. [screenshot] [2]

(e) What is the throughput (bytes transferred per unit time) of the TCP connection? Explain how you calculated this value. [screenshot] 2

## Part II: UDP

**Q.2** Load the file `"udp-1.pcapng"` in Wireshark and answer the following questions.

(a) Find a UDP packet in the trace file and describe the name and length (in bytes) of each of the UDP header fields. 2

(b) Using the statistics feature of Wireshark, determine the percentage of IPv4 UDP packets in the trace. 2

(c) Select a UDP packet in the trace file and expand the "User Datagram Protocol" header. What does the value of the "Length" field represent? (You may consult the textbook for this answer). Verify your claim by showing the relevant details of your selected UDP packet. [screenshot] 2

(d) What are the source port and the length of the first UDP packet in the trace file? [screenshot] 2
What is the largest possible source port number?

(e) What is the protocol number for UDP? Give your answer in both hexadecimal and decimal notations along with a screenshot of Wireshark showing those values. [screenshot] 2

## Part III: IP

**Q.3** Load the file `"ip-1.pcapng"` in Wireshark and answer the following questions. The trace was generated while executing three `traceroute` commands with three different UDP datagram sizes (56, 2000, and 3000 bytes) on a computer with an IP address of `10.63.7.60`. The IP address of the target host was `103.74.84.13`.

(a) Select any of the ICMP echo packets from the trace. Within the IPv4 packet header, what is the value of the upper layer protocol field? 1

(b) What are the values in the "Identification" and the "TTL" fields of the first ICMP packet (ping reply) received by `10.63.7.60`? 2

(c) Select any of the UDP packets generated from the second `traceroute` command. How many bytes are in the IP header? How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes. [screenshot] [2]

(d) Observe the UDP packets generated from the first `traceroute` command. Which fields in the IP datagram *always* change from one datagram to the next within this series of messages sent by `10.63.7.60`? [screenshot] [2]

(e) Do the values of the "Identification" and the "TTL" fields remain unchanged for all of the ICMP TTL-exceeded replies sent to your computer (IP `10.63.7.60`) by the nearest (first hop) router? Why or why not? [screenshot] [3]

**Report Submission on Canvas**

☞ Upload a single PDF/DOC file to Canvas. All screenshots should be properly organized within the PDF/DOC.

☞ Separate screenshots (image files) or ZIP files are NOT acceptable.